

CSE587: Networks for AI/ML Systems

QUIZ 3 (30 mins) [Total: points]

Sep 30, 2025

Q.1. Suppose the **TCP congestion control** algorithm starts with the **Slow Start** phase with a **window size of 1MSS** and the slow start threshold (ssthresh) is 64000 bytes. If the receiver window size is 16000 bytes and the maximum segment size is 1000 bytes, then **after how many RTT sender will send full window (i.e., 16000 bytes)**? Assume no timeouts, no packet losses, and no errors. Justify. [3]

Ans.

After 5 RTT

1 segment size, MSS=1000 bytes; Window size = 16000 bytes = 16 MSS (i.e., 16 segments)

Slow start: After each RTT double the congestion window [1]

First RTT: Send 1 segment

Second RTT: Send 2 segments (total = 3 segments)

Third RTT: Send 4 segments (total = 7 segments)

Fourth RTT: Send 8 segments (total = 15 segments)

Fifth RTT: Send 1 more segment (to make the total 16 segments) [2] → Even if you write 16 segments instead of 1 segment here, it is fine

Q.2. The routing table of a router is shown below: [1+1+1]

Destination	Subnet mask	Interface
125.12.41.0	255.255.128.0	1
125.12.32.0	255.255.224.0	2
180.128.112.0	255.255.240.0	3
180.128.112.0	255.255.248.0	4
180.128.112.0	255.255.255.0	5
Default		0

On which interface will the router forward packets addressed to the following destinations? Explain how you obtain the answer. **Note that you will NOT be awarded partial points for correct answers without explanation.**

(a) 125.12.41.10

(b) 180.128.119.31

(c) 10.129.116.10

Ans:

(a) 2

$125.12.41.10 \wedge 255.255.128.0 = 125.12.0.0 \Rightarrow$ does not match the dest network "125.12.41.0" ;

$125.12.41.10 \wedge 255.255.224.0 = 125.12.32.0 \Rightarrow$ matches the dest network "125.12.32.0"

(b) 4

$180.128.119.31 \wedge 255.255.240.0 = 180.128.112.0 \Rightarrow$ matches the dest network "180.128.112.0"

$180.128.119.31 \wedge 255.255.248.0 = 180.128.112.0 \Rightarrow$ matches the dest network "180.128.112.0"

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$180.128.119.31 \wedge 255.255.255.0 = 180.128.119.0 \Rightarrow$ doesn't match dest network "180.128.112.0"
Longest prefix that matches is "255.255.248.0"; the packet should be forwarded to interface 4

(c) 0

The destination IP address will not match with any routing table entry. Therefore, the packet is forwarded via the default interface, "0".

Q.3. How does TCP identify that congestion has occurred? Why does this congestion signal fail in data center networks that run HPC (High Performance Compute) or distributed ML applications? **[2+2]**

Ans.

TCP uses "Timeout" at the sender process/application as the congestion signal. Timeout value is derived based on the RTT value. Since RTT value is in microseconds in data centers, TCP's reaction to congestion is slow due to the slow software-based kernel network stack. Therefore, we rely on network-assisted or hybrid mechanisms to signal congestion.

Q.4. DCTCP relies on switches to mark packets with ECN when the queue size crosses a threshold. What is one of the important limitations of the ECN-based approach? **[2]**

Ans.

Selecting an appropriate queue length threshold and maintaining the configurations as line speeds and buffer sizes vary across network switches is challenging at scale. Also, ECN is a binary (not a multi-bit) signal, i.e., it indicates a congestion signal but fails to indicate the extent of congestion.

Q.5. What is the main limitation of RTT-based congestion control, where the receiver sends the ECN packet to the sender, and the sender monitors the RTT as well to indicate the extent of congestion? **[2]**

Ans.

RTT measurements lump queueing in both directions along the network path. This may confuse reverse path congestion experienced by ACKs with forward path congestion experienced by data packets. One simple fix is to send ACKs with higher priority, so that they do not incur significant queueing delay.
